

Oluwatobi Melvyn Mayungbo

AI Engineer, LLM and Retrieval Systems

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SUMMARY

AI engineer with a B.Sc. in Statistics focused on evaluation. Built a retrieval assistant scored on a labelled golden set with retrieval and generation measured separately, and traced a weak judge result to the judge itself. Runs experiments with bootstrap intervals, power analysis and CUPED, and ships services with Docker and CI.

SKILLS

AI and LLM Systems: LLM Evaluation and Guardrails, Retrieval-Augmented Generation (RAG), Citation Verification, Model Context Protocol (MCP), Hybrid Search (BM25, Dense Embeddings, RRF), Prompt Engineering, Hugging Face, ONNX Runtime

Statistics and Causal Inference: Experiment Design, Power Analysis, CUPED, Statistical Inference, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

Engineering and MLOps: Python, Prefect, FastAPI, Docker, CI/CD (GitHub Actions), Pytest, SQL, DuckDB, PostgreSQL, Redis, GCP Cloud Run, AWS ECS Fargate, Railway, dbt-core, MLflow, Evidently, Git, Streamlit, React

Machine Learning: PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

Payments and Risk: Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

PROJECTS

Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run

- Ran retrieval and generation evaluation as separate metrics to localise failure by component; traced a 0.40 refusal-correctness result to a lexical-overlap limitation in the offline stub judge and flagged re-validation against a live provider as the next step instead of overstating readiness.
- Designed generation to fail safely under uncertainty: a sentinel-token refusal mechanism enforced across the prompt, output guardrail and evaluation harness, plus post-generation citation verification that strips any marker the model cited but never retrieved.
- Built and deployed a regulatory question-answering system over central bank (CBN) circulars and the NDPA, combining sparse (BM25) and dense (ONNX-served MiniLM) retrieval fused via reciprocal rank fusion, reaching 0.96 recall@5 and 0.842 MRR on a 50-question labelled golden set with zero retrieval misses in the top 10.

Uplift Modelling for Campaign Targeting | [GitHub](#)

Tech: Python, scikit-learn, LightGBM, EconML, pytest, GitHub Actions

- Added a power and MDE calculator and a CUPED variance-reduction demo; 135 tests on synthetic data with CI on Python 3.10 to 3.12.
- Evaluated with Qini curves, AUUC, decile tables and bootstrap intervals; uplift targeting was worth about \$1,390 against about \$3,170 for raising the budget, and a failed placebo test on the primary campaign was reported openly.

Self-Operating Demand Forecasting Platform | [GitHub](#)

Tech: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions

- Built an end-to-end hourly demand forecasting system for NYC taxi zones, producing 24-hour quantile forecasts through an orchestrated Prefect pipeline spanning ingestion, dbt/DuckDB transformations, backtesting, drift monitoring and conditional retraining.
- Validated with rolling-origin backtesting across 6 expanding-window folds against seasonal-naive and historical-mean baselines, reporting MASE and pinball loss, with automated leakage tests that fail the build if any feature reads past the forecast origin.

Sequence-Based Transaction Fraud Detection System | [GitHub](#)

Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Benchmarked all architectures against a strong LightGBM baseline on engineered velocity and deviation features, reporting PR-AUC and precision at fixed review capacity (top 0.1%, 0.5%, 1%) across 3 seeds, with a per-attack-type breakdown covering card testing, account takeover and merchant compromise; sequence models cut expected cost per transaction 27% under an explicit cost model.
- Deployed real-time scoring by exporting to ONNX Runtime (2.9× faster than eager PyTorch at batch size 1) and serving via FastAPI, reaching 5.1 ms p99 end-to-end latency against a 50 ms budget, with append-only per-decision audit logging and attention-based attribution of the prior transactions driving each flag.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

Internal Auditor, NISER May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

B.Sc. Statistics, University of Ibadan January 2021 to March 2025

Professional Diploma in Artificial Intelligence, SQI College of ICT July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action